Statistical Methods in Artificial Intelligence
Assignment 1

Spring 2026 · Instructor: Prof. C. V. Jawahar

General Instructions

G1. Academic Integrity. All work must be your own. You may discuss ideas at a high level with classmates, but all code, derivations, plots, and written analysis must be individually produced. Copying code or text from any source (including AI assistants) without attribution is a violation of the honour code.

G2. Submission Format. Submit a single ZIP file containing: (i) a PDF report with all answers, plots, tables, and analysis; (ii) all source code files (well-commented `.py` or `.ipynb`).

G3. Reproducibility. Fix and report all random seeds. A grader must be able to re-run your code and obtain the same results.

G4. Allowed Libraries. Unless stated otherwise, you may use only `numpy`, `matplotlib`/`seaborn`, and `pandas`. Using `sklearn` or any external ML library for core algorithms is **prohibited** unless explicitly permitted in a question.

G5. Late Policy. Submissions after the deadline receive zero marks.

G6. Variant Assignment. Each student is assigned one variant per question (see instructions within each question). Solving the wrong variant will receive zero marks for that question.

Marks Summary

Question	Topic	Marks
Q1 (one variant)	Variant A: Bias–Variance Tradeoff & Regularisation	35
	OR Variant B: Face Recognition via Subspace Methods	35
	OR Variant C: Clustering Handwritten Digits	35
Q2	Data Normalisation & KNN Classification	30
Q3 (one variant)	Variant A: Linear Regression Under Noise & Outliers	35
	OR Variant B: SVM for Text Classification	35
Q4 (individual)	Concept Deep-Dive: Research & Peer-Learning Synthesis	15
	Total	**115**

1 Question 1 [35 marks]

This question is an expanded version of the regular class work. A bit longer than the regular one. You will complete **exactly one** of the three variants below. All variants carry equal marks (35). Three variants are - (i)Bias-Variace Tradeoff, (ii)Face Recognition via Subspace Methods and (iii) Clustering Handwritten digits without explicit labels.

To select your variant, compute your roll number % 3. If the result is 0, pick (i)Bias-Variace Tradeoff; if the result is 1, pick ii)Face Recognition via Subspace Methods; and (iii) if the result is 2, pick (iii) Clustering Handwritten digits without explicit labels. In case of any doubts, contact any of the TAs.

Follow the instructions in the jupyter notebook carefully.

2 Question 2 — Data Normalisation and KNN Classification [30 marks]

Context

Preprocessing is not a mere formality—the choice of normalisation can make or break a distance-based classifier. In this question you will implement 20 normalisation methods from scratch, apply them to a personalised dataset fetched via an API, and rigorously evaluate their impact on KNN accuracy. The dataset variant you receive depends on your roll number, ensuring every student faces a unique data challenge.

2.1 The Normalisation Library [10 marks]

Implement all 20 methods listed in Appendix A using **only** `numpy` and `math`.

Marking

Each correctly implemented method: 0.5 marks. Total: $20 \times 0.5 = 10$ marks. Partial credit: correct formula but buggy edge-case handling receives 0.25 marks.

2.2 Dataset Collection [2 marks]

You do not receive a CSV file. Instead, you must fetch your data point-by-point from the provided API (see Appendix C).

2.2.1. Write a script that polls the API and collects $N \geq 1{,}000$ samples into a Pandas DataFrame. **[1 mark]**

2.2.2. Your scenario is determined by: Scenario ID = Roll Number mod 8. See Appendix B for scenario descriptions. Report your roll number, scenario ID, and a brief description of the scenario in your report. **[1 mark]**

2.3 Experimental Protocol [6 marks]

2.3.1. **Baseline** (Raw Data): Run KNN on the raw, unnormalised data. Report accuracy. **[1 mark]**

2.3.2. **Train/Test Split**: 80%/20%, fixed random seed. **[0.5 marks]**

2.3.3. Fit–Transform Rule: Compute all normalisation statistics (min, max, mean, std, quantiles, etc.) using **only the training set**. Apply the stored statistics to transform both the training and test sets. **[1.5 marks]**

2.3.4. Hyperparameter Tuning: For each of the 21 configurations (Raw + 20 methods), perform 5-fold cross-validation on the training set to select the best $k \in \{1, 2, \ldots, 30\}$ for the KNN classifier using Euclidean distance. **[2 marks]**

2.3.5. Final Evaluation: Train with the optimal k on the full training set; evaluate on the test set. Report test accuracy and macro F1-score. **[1 mark]**

2.4 Deliverables and Analysis [12 marks]

2.4.1 Master Results Table [4 marks]

Submit a single table (one row per method) containing: Method Name, Best k, Test Accuracy, Macro F1-Score, and Rank (by test accuracy). See the template in Appendix D.

2.4.2 Visual Analysis [4 marks]

V.1. Feature Distribution Plot [2 marks]: Select one feature. Plot its histogram/KDE for (a) Raw Data, (b) the Top-performing method, and (c) the Worst-performing method. Annotate each sub-plot with the method name.

V.2. Accuracy vs. k Plot [2 marks]: A line graph with k on the x-axis and cross-validation accuracy on the y-axis. Plot three lines: Raw Data, Best Method, Worst Method.

3 Question 3 [35 marks]

3.1 Linear Regression Under Noise, Correlation, and Outliers

Context

Real-world data rarely satisfies textbook assumptions: features may be correlated, observations may contain outliers, and numerical conditioning can cause instability. In this variant you will implement multiple regression algorithms from scratch and systematically study their behaviour under adverse conditions.

Expected Effort

Estimated time: 8–10 hours. Deliverables: code (from-scratch implementations), 6–8 figures, and written analysis ($\sim$2,500 words total).

Allowed Libraries

`numpy`, `matplotlib/seaborn` only. No `sklearn` or external regression libraries.

3.1.1 Part A — Dataset Construction [5 marks]

Construct three synthetic datasets. All design choices (sample size $\geq$ 10,000, coefficients, noise variance, etc.) are yours, but must be clearly documented and justified. Fix and report all random seeds.

A.1. Clean Dataset: Generate 1D data from $y = ax + b + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Report your choices of a, b, σ, sample size, and distribution of x. **[1.5 marks]**

A.2. Correlated Feature Dataset: Create a multi-dimensional dataset (≥ 2 features) where features are highly correlated (e.g., one feature is a near-linear function of another plus small noise). Compute and report the eigenvalues and condition number of $X^T X$. **[2 marks]**

A.3. Outlier Dataset: Starting from the clean dataset, introduce *structured* outliers in $\geq 10\%$ of the samples (e.g., large additive deviations to y). Describe the corruption process precisely. **[1.5 marks]**

3.1.2 Part B — From-Scratch Implementations [6 marks]

Implement each of the following. Show the key equations in your report.

B.1. Closed-form ordinary least squares (OLS). **[1 mark]**

B.2. Gradient descent for OLS (MSE objective). **[1 mark]**

B.3. Ridge regression—both closed-form and gradient descent. **[1.5 marks]**

B.4. Lasso regression via sub-gradient descent. **[1.5 marks]**

B.5. Weighted least squares (closed-form). **[1 mark]**

3.1.3 Part C — Gradient Descent Analysis [6 marks]

Using the clean dataset:

C.1. Train with at least three learning rates. Plot loss vs. iteration for each. Show one rate that converges, one that oscillates, and one that diverges. **[2.5 marks]**

C.2. Relate convergence behaviour to the eigenvalues of $X^T X$. Specifically, state and verify the condition $\eta < 2/\lambda_{\max}$ for convergence. **[2 marks]**

C.3. Written (5–8 sentences): Explain theoretically why the maximum eigenvalue of $X^T X$ governs the convergence bound. **[1.5 marks]**

3.1.4 Part D — Ill-Conditioning and Ridge Regression [6 marks]

Using the correlated feature dataset:

D.1. Report eigenvalues and condition number of $X^T X$. **[1 mark]**

D.2. Train OLS and Ridge regression ($\lambda \in \{0.001, 0.01, 0.1, 1, 10\}$) using $\mathbf{w} = (X^T X + \lambda I)^{-1} X^T y$. Plot: (i) weight magnitudes vs. λ, and (ii) test MSE vs. λ. **[2.5 marks]**

D.3. Written (5–8 sentences): Explain how adding λI to $X^T X$ improves the condition number and stabilises the solution. **[2.5 marks]**

3.1.5 Part E — Outlier Stress Test [5 marks]

Using the outlier dataset:

E.1. Train OLS (closed-form), OLS (gradient descent), Ridge, and Lasso. Plot all regression lines on a single scatter plot. Report training MSE and learned weights for each. **[2.5 marks]**

E.2. Written (5–8 sentences): Analyse which method is most and least robust to outliers, and why. **[2.5 marks]**

3.1.6 Part F — Iteratively Reweighted Least Squares [7 marks]

Consider the weighted objective: $J = \sum_{i=1}^{N} \gamma_i (y_i - \mathbf{w}^T \mathbf{x}_i)^2$. The closed-form solution is $\mathbf{w} = (X^T \Gamma X)^{-1} X^T \Gamma y$ where $\Gamma = \text{diag}(\gamma_1, \ldots, \gamma_N)$.

F.1. Implement the following iterative scheme on the outlier dataset: **[3 marks]**

(a) Initialise $\gamma_i = 1$ for all i.

(b) Compute $\mathbf{w}$ using the weighted closed-form.

(c) Update: $\gamma_i = \frac{1}{1+|y_i - \mathbf{w}^T \mathbf{x}_i|}$.

(d) Repeat (b)–(c) until convergence (or a maximum of 50 iterations).

F.2. Plot the evolution of the regression line and the distribution of weights γ_i across iterations (at least iterations 1, 5, 10, and the final iteration). **[2 marks]**

F.3. **Written (5–8 sentences):** Does the method converge? Does it successfully down-weight outliers? Compare the final regression line to OLS and Ridge. **[2 marks]**

4 Question 4 — Concept Deep-Dive [15 marks]

Context

A hallmark of genuine understanding is the ability to *explain* a concept clearly to someone else. In this question you will research a single focused topic from the course, synthesise what you learn, and produce a short document that could serve as a study resource for your classmates.

Expected Effort

Estimated time: 4–5 hours (research, writing, and diagram preparation).
Deliverable: a single PDF document of **3–5 paragraphs** (maximum 2 A4 pages), uploaded via the course portal.

Portal-Based Topic Assignment

When you sign in to the course portal, **one topic will be randomly assigned to you** from the bank of 300 topics. Each student will receive a *unique* topic. You must complete the topic assigned to you—submitting work on a different topic will receive zero marks. Upload your completed PDF on the portal itself.

Task Description

Your goal is to research your assigned topic using the lecture material, textbooks, and reputable web resources, and then write a clear, self-contained explanation that a fellow SMAI student could use as a study guide.

Your document must include:

(a) **Conceptual Explanation** — Describe the idea in your own words. Build intuition *before* introducing formalism. **[4 marks]**

(b) **Mathematical / Algorithmic Core** — Include at least one of the following: a derivation, an algorithm walkthrough, or a numerical dry-run that demonstrates the mechanics. **[4 marks]**

(c) **Visual Element** — Include at least one diagram, plot, table, or worked example that aids understanding. **[3 marks]**

(d) **Connections** — Briefly relate your topic to at least one other concept from the course. **[2 marks]**

(e) **References** — Cite the sources you consulted (lecture slides, textbook chapters, URLs). **[2 marks]**

Formatting guidelines

- Title your document with your assigned topic number and title.
- Use clear section headings (e.g., *Intuition*, *Derivation*, *Example*, *Connections*).
- Maximum **2 A4 pages**, font size $\geq 11\,\text{pt}$, reasonable margins.
- Submit as a **PDF** via the course portal.

A Normalisation Methods for Q2

Implement all 20 methods below. Each method should expose a `fit(X_train)` method (to compute and store statistics) and a `transform(X)` method (to apply the transformation).

A. Linear Scaling (Bounding the Range)

1. **Min–Max Scaling:** $X_{\text{new}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$. Bounds output to $[0, 1]$.
2. **Mean Normalisation:** $X_{\text{new}} = \frac{X - \mu}{X_{\max} - X_{\min}}$. Centres at 0.
3. **Maximum Absolute Scaling:** $X_{\text{new}} = \frac{X}{\max(|X|)}$. Preserves sparsity.
4. **Decimal Scaling:** $X_{\text{new}} = \frac{X}{10^j}$ where j is the smallest integer such that $\max(|X_{\text{new}}|) < 1$.
5. **Robust Scaling (IQR):** $X_{\text{new}} = \frac{X - Q_2}{Q_3 - Q_1}$. Uses median and inter-quartile range.

B. Statistical Normalisation (Centering & Variance)

6. **Z-Score Standardisation:** $X_{\text{new}} = \frac{X - \mu}{\sigma}$. Mean $= 0$, Std $= 1$.
7. **Modified Z-Score:** $X_{\text{new}} = \frac{0.6745\,(X - \tilde{x})}{\text{MAD}}$ where MAD is Median Absolute Deviation.
8. **Pareto Scaling:** $X_{\text{new}} = \frac{X - \mu}{\sqrt{\sigma}}$. Scales by square root of SD.

C. Non-Linear Transformations (Shape Changing)

9. **Log Transformation:** $X_{\text{new}} = \log(X + c)$. Determine c to handle non-positive values.
10. **Reciprocal Transformation:** $X_{\text{new}} = 1/X$. Handle division by zero.
11. **Square Root Transformation:** $X_{\text{new}} = \sqrt{X}$. Handle negative values.
12. **Box–Cox ($\lambda = 0.5$):** $X_{\text{new}} = \frac{X^{\lambda} - 1}{\lambda}$. Requires $X > 0$.
13. **Yeo–Johnson ($\lambda = 0.5$):** Handles negative values in power transforms.
14. **Hyperbolic Tangent:** $X_{\text{new}} = \tanh(X)$. Squashes to $[-1, 1]$.
15. **Sigmoid / Logistic:** $X_{\text{new}} = \frac{1}{1 + e^{-X}}$. Squashes to $(0, 1)$.

D. Vector Normalisation (Row-wise)

16. **L1 Normalisation:** $X_{\text{new}} = X / \|X\|_1$. Row sums to 1 in absolute value.
17. **L2 Normalisation:** $X_{\text{new}} = X / \|X\|_2$. Unit vector scaling.
18. **Softmax Scaling:** $X_{\text{new},i} = e^{X_i} / \sum_j e^{X_j}$. Converts vector to probability distribution.

E. Distribution Mapping

19. **Quantile Normalisation (Uniform):** Maps data ranks to a uniform distribution on $[0, 1]$.
20. **Rank–Gauss:** Maps ranks to a Gaussian distribution via the inverse CDF.

B Data Scenarios for Q2

Your scenario is determined by: Scenario ID = Roll Number mod 8.

ID	Name	Description
0	The Skewed Reality	Features follow a Log-Normal or Power Law distribution. Most values are small, but "whale" outliers exist. Standard Z-Scores will be distorted by the long tail.
1	The Broken Sensor	Gaussian data contaminated with massive random noise (±10,000). Mean and standard deviation are unreliable estimators.
2	Apples & Oranges	Features have vastly different scales (e.g., Age vs. Salary). One feature dominates Euclidean distance, rendering others irrelevant.
3	The Direction Vector	High-dimensional data where angle matters more than magnitude. Common in text vectors. Centering may destroy sparsity or geometric meaning.
4	The Double Peak	Bimodal distribution (camel-back shape). A single mean falls in the valley where no data exists, misleading the classifier.
5	The Hard Limit	Data bunched against a maximum or minimum (censored data). Linear scaling crowds critical high-value points.
6	The Negative Skew	Long tail to the left with negative values. Log transform is undefined for negatives; advanced power transforms required.
7	The Sparse Manhattan	Integer counts with many zeros. In this space, L2 norm often underperforms L1 or presence/absence scaling.

C Data Generation API for Q2

Item	Details
Endpoint	`get_data(roll_number, index)`
Parameters	`roll_number`: your university ID (integer). `index`: integer from 0 to $N_{\max}$ (e.g., 2,000).
Return	JSON object: {`"features": [...], "label": 0`}
Determinism	Same parameters always yield the same result.
Bounds	If `index` exceeds the dataset limit, returns `null`.

Batch Collection Required

You cannot normalise a single data point in isolation—normalisation requires population statistics. First collect all samples into a DataFrame, *then* split into train/test, *then* normalise.

D Master Results Table Template for Q2

Method	**Best k**	**Test Accuracy**	**Macro F1**	**Rank**
Raw Data (baseline)	—	—	—	—
1. Min–Max Scaling				
2. Mean Normalisation				
3. Max Absolute Scaling				
...	...	...	...	...
20. Rank–Gauss				

E Marking Rubric Summary

Criterion	**Weight**	**Description**
Correctness	40%	Code produces correct results; derivations are mathematically sound; plots are accurate.
Analysis Quality	25%	Written answers demonstrate understanding (not just surface-level descriptions); connections are drawn between theory and experiments.
Presentation	15%	Report is well-organised; figures have titles, axis labels, and legends; tables are properly formatted; code is well-commented.
Reproducibility	10%	Random seeds are fixed and reported; code runs without errors; results match the report.
Effort & Exploration	10%	Evidence of experimentation beyond minimum requirements (e.g., additional hyperparameter sweeps, extra visualisations, insightful edge-case analysis).